

GRAVITYSORT · DESIGN DECISIONS REGISTER

GravitySort design decisions register

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DOCUMENT CONTROL

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Project TRL	TRL 3 Proof of concept		
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Source	docs/06-design-decisions.md @ uncommitted		

REVISION HISTORY

Version	Date	Author	Change
0.1	2026-10-01	Amish Chadha	Register opened with the build plan; open decisions from the review notes, the decision records and the design for construction

GravitySort design decisions register

Every design decision still to be made, and every decision made, in one place. Each decision is argued in full in its decision record under `docs/decisions/` ; this register is the index Amish works from. The build plan (`docs/05-build-plan.md` , GVS-BLD-001) describes the design as it stands and does not list open decisions.

■ Open decisions

#	Decision needed	Options	Recommendation	Affects in the build	Source
1	Accept the design for construction	Accept the changes P1 to P15 as made; accept with changes; return to the concept layout	Accept: every change keeps what the machine does and all 102 model checks pass	The whole build plan	GVS-DDR-003, Table 1
2	R11 total mass: the constructable design is 95.7 kg against 80 kg, with every load under 30 kg (heaviest 25.2 kg)	(a) restate R11's total as 100 kg or less, keeping loads of 30 kg or less; (b) look for about 16 kg of savings, then re-estimate; (c) keep 80 kg, R11 not met	(a), because the per-load limit decides whether two people can carry it; try the savings in (b) at TRL 4	None now; (b) would change plates, hopper, guards and table base	GVS-DDR-003, A1; GVS-CAL-001 v0.4 section 11
3	How the table gets its asymmetric stroke	(a) adjustable rubber bump stop at the return end; (b) toggle head; (c) spring return with a cam	(a) for the prototype; (b) if the first test shows it is not enough	Table head and the frame's table end (section 3.16 and step 16 of the plan)	GVS-DDR-003, A2; GVS-PRC-001 item 15
4	Pedal position for riders of different sizes	(a) keep the bicycle-style seat, about 910 to 960 mm up, seat post adjustable; (b) a recumbent seat further back	(a); check with two riders at TRL 4	Pedal outrigger (section 3.2)	GVS-DDR-003, A3
5	Head for the fluidization supply	(a) a 1.7 m tank post (8.1 kPa at the union); (b) keep the 1.25 m post and add a small pump	None yet	Tank post height or a pump on the water line	GVS-DDR-001, item 11b
6	Water from the settling pond at pedal-only sites (about 1.19 m ³ /h, 5.5 W hydraulic)	Hand pump; a second rider on a pump; gravity supply from upstream	None yet	Not part of the machine; the header tank is filled by it	GVS-DDR-001, item 12
7	R2 throughput margin (1.55 t per shift after flush stops)	Raise the design feed to about 210 kg/h; restate R2 as 200 kg/h of feed time	None yet	None (operating figure)	GVS-DDR-001, item 13
8	First co-design partner and country	A partner in a country with a mercury ban in force (Colombia) or with a Minamata action plan and a planetGOLD programme	None yet (the screening criterion is decided)	None until a field trial	GVS-DDR-001, item 8b
9	Containment of a liner fragment by the tub and lid (61 J at 1,200 rpm)	(a) check by calculation before any spin test; (b) add a steel band round the tub at the bowl lip height	(a), then (b) only if the check fails	Splash tub and lid	GVS-CAL-001 section 10; review note 2026-09-25
10	A sight window in the lid	(a) keep the plain 10 mm HDPE lid; (b) a window of 6	(a)	Lid guard (section 3.11)	

#	Decision needed	Options	Recommendation	Affects in the build	Source
		mm polycarbonate or more after the containment check			Review note 2026-09-26, item 4
11	Concentrate security in practice	Padlock on the box and container; sealed container; a two-person rule	Decide with the partner	Concentrate box and flush container	GVS-PRC-001, open questions
12	Renders: the cranks turned 100 degrees from top dead centre so the seated figure reads as pedalling	Accept as a render pose only; render with the model's crank angle	Accept as a render pose only	Renders only	Review note 2026-09-26, item 1

■ To confirm when parts are bought

#	What to confirm	Why it matters	Source
1	The two flanged bearing units (UCF205 class): 95 mm square flange, bolts 70 mm apart, 25 mm bore with set screws	They set the holes in the two bearing plates	GVS-DDR-003, P2
2	The shaft flange hub: 25 mm bore, 80 mm flange, four holes on a 60 mm circle, room for a 6 mm cross pin	It joins the bowl and jacket to the spindle	GVS-DDR-003, P4
3	The rotary union: 1/2 in BSP on the shaft side, side water port, seal drag near 0.10 N m	Pedal power (R6) assumes this drag; the nipple welded into the spindle must match its thread	GVS-CAL-001 sections 7 and 10
4	The right-angle 1:1 gearbox: through input shaft about 33 mm above its feet, feet about 100 x 80 mm, output shaft long enough for the drive pulley 85 mm below the feet	It sets the jackshaft height and the member spacing	GVS-DDR-003, P9; GVS-PRC-001 open questions
5	The four 20 mm pillow blocks (UCP204 class, centre height 33.3 mm, bolts 95 mm apart)	Two set the jackshaft height on the gearbox members; two carry the table head shaft	GVS-DDR-003, P8 and P14
6	The two rubber pipe grommets for 63 mm and 75 mm pipe that seal in a 6 mm curved wall	They seal the standpipe and tailings pipe without welding	GVS-DDR-003, P6
7	The 430 mm HDPE drum: flat floor, 6 mm wall, food-grade or clean	It is the splash tub	GVS-DDR-003, P6
8	The used bicycle parts: a bottom bracket shell that can be cut from a scrap frame and welded, crank axle length, a 27.2 mm seat post	They set the outrigger's pedal post and seat tube	GVS-DDR-003, P8
9	The disc caliper's mount (post mount or the older standard) and the disc's 6-bolt pattern	They set the caliper bracket holes and the brake flange	GVS-DDR-003, P15
10	MotionCore: the reference hub motor's no-load speed (assumed 250 rpm), a sprocket on its disc mount, axle length for dropouts 76 mm apart inside, a speed limit that maps to 900 rpm at the spindle, a brake input for the lid switch	They set the 1.2:1 step-up, the cradle and the motor speed cap	GVS-DDR-002, item 14; review note 2026-09-25

■ Value engineering

Value-engineering target: USD 455 (a hypothetical control target, not a limit; Amish, 2026-10-01). Estimated cost of the constructable design: USD 558 (USD 103 over the target), excluding the MotionCore kit, reference motor and battery. Main cost drivers and savings worth trying:

- **Main cost drivers.** The jackshaft, gearbox, chains and pulleys (\$60); the jacket, hub and rotary union (\$55); the table stand and head (\$52); the bowl (\$46); the frame (\$43); the water supply (\$41). Making the design buildable added \$103, mostly the table head and tensioner, more frame tube and plate, and the tub fittings (GVS-DDR-003).
- **A quarter-turn belt instead of the bevel gearbox,** about \$25 less (GVS-DDR-001, item 1). Worth a test, since a used gearbox is also the hardest part to find.

- **Used bearing units and pillow blocks** from scrap farm or factory machinery, about \$10 to \$15 less across the six bearings.
- **Run the table head shaft in two plain bronze bushes** in the head plate instead of two pillow blocks, about \$8 less; the shaft turns slowly (about 300 rpm).
- **Take the table wash water from the main hose** with a tee at the rotameter instead of a separate branch and valve, about \$4 less.
- **Buy steel as offcuts** and use one plate thickness (5 mm) throughout, a few dollars and a little mass.
- Not recommended: the non-fluidized bowl (about \$45 less) or building the table later (about \$100 deferred), because both change what the machine does.

■ Decisions made

Date	Decision	Decided by	Record
2026-09-25	TRL 2 review items: two stages, fluidized bowl with rotary union, cast PU liner, pedal baseline with the MotionCore option and one time-shared drive, battery left to the user, borax smelting outside the machine, partner screening criterion	Amish: "i accept all your recommendations, go with them across all repos."	GVS-DDR-001, GVS-DDR-002
2026-09-25	Budget from \$350 to \$450; R12 restated for the pedal drive; frame in 25 x 25 x 1.5 mm tube; 3/4 in fluidization hose and graded holes; disc brake with parking latch and lid interlock pin, speed display, 1.2:1 motor step-up, MotionCore lid switch	Amish, same instruction	GVS-DDR-002, items 1, 9, 10, 11a and 14
2026-09-26	Budget set to \$455 to cover the priced BOM	Amish: "i approve all the budget items."	GVS-DDR-002 v0.2
2026-09-26	GravitySort chosen for the first batch of product renders	Amish	docs/ REVIEW.md , session of 2026-09-26
2026-09-30	Make the design physically buildable while drawing the build plan; the changes are recorded in GVS-DDR-003 and are open for review (open decision 1)	Amish: "If you are realising that the design cannot be built as per concept - fix the design assumptions to match and be physically feasible as you draw the illustrations."	GVS-DDR-003
2026-09-30	Open decisions are kept out of the build plan, in this register	Amish: "don't log outstanding decisions in this build plan - that is not the place for it. that should be in a separate design document logged and named as such"	.kit/ STANDARDS.md section 18
2026-10-01	The budget is a value-engineering target, not a limit; cost is reported against it	Amish: "the budgets are a hypothetical control target to ensure we are thinking along a value engineering lens. its ok to ensure wording reflects that the hypothesis budget was x - the real cost being accrued is y"	.kit/ STANDARDS.md section 18